

Sources of between-run variability in language-model vocabulary learning: a comparison to variability in children

Background and significance

Children's early vocabulary grows very rapidly, with large variation between children. Some children know a few words by their first birthday; others know hundreds. This variability is one of the most robust features of early language development, and understanding where it comes from has theoretical, practical, and clinical importance: it bears on the mechanisms of word learning, it predicts later academic achievement, and it shapes how we identify and support children with language delays.

A common view in both developmental psychology and machine learning treats word learning as a process of accumulation — more exposure to language yields more vocabulary in a predictable, additive way. This view is a conceptual bridge to large language models, whose learning behavior is well-described by smooth scaling laws in the amount of training data they see. If children are pure accumulators, then modern language models may be reasonable computational accounts of early language acquisition; if they are not, then the apparent similarity is superficial.

Our prior work suggests the second answer. Across large samples of children's vocabulary growth, we find that vocabulary acquisition *accelerates* over development in a way that pure accumulation cannot explain, and that variation in how much language children hear accounts for only about 10% of the variability between children. Most between-child variability in vocabulary growth comes from sources other than input quantity.

Language models, in contrast, *do* appear to behave like pure accumulators: their learning is well-fit by smooth scaling relationships. But there is a missing piece in this comparison. For children, we measure variability *between individuals*. For language models, almost nobody has measured the corresponding quantity. The typical evaluation reports performance for a single trained model (or averages over a few seeds), without asking how much two language models trained under nominally identical conditions actually differ in what they have learned. Without that measurement, the apparent contrast between "variable children" and "deterministic LMs" cannot be sharpened into a real claim.

This project measures between-LM variability in vocabulary learning, decomposes it into contributions from different sources of variation, and compares it directly to the between-child variability we measure in CDI data (over 5,000 children).

Building on existing infrastructure

A prior run on Sherlock has already trained 3 GPT-2-small models on the same set of child-directed speech (a filtered 25M-token English subset of the CHILDES corpus). All three models gave nearly identical vocabulary-learning curves on the 611 words in the MacArthur-Bates

Communicative Development Inventory. This tells us that seed-only variability at full data scale is small.

The next question is whether *other* sources of variation — particularly variation in the data each model sees, and variation in architecture — produce larger differences between models, and how those differences compare to the between-child variability we see in CDI data. Answering that requires training a population of models, not just a few seeds.

The training and evaluation pipeline is already built, runs end-to-end on Sherlock, and produced the 3 completed seeds. The proposed Marlowe project ports it to Marlowe and scales it out.

Phase 1: portability pilot (2 runs)

The first phase trains **2 GPT-2-small models** on two non-overlapping 10M-token chunks of CHILDES (the maximum number of fully disjoint ~10M chunks the corpus permits). Same architecture, same random seed, same training recipe — the only intentional difference is the data each model sees.

This 2-run pilot has two purposes:

1. **Validate the Sherlock-to-Marlowe port.** Confirm that the training pipeline reproduces our existing 3-seed CHILDES results on Marlowe before scaling to 90 runs.
2. **Decide the data substrate for Phase 2.** If the two 10M-CHILDES models give noticeably different vocabulary-learning curves, the 90-run study's data-variation axis uses CHILDES at a slightly smaller scale (to support more disjoint subsamples). If they look identical, the data axis uses TinyDialogues (Feng et al., 2024) — a 200M-word synthetic child-directed-speech corpus that permits many non-overlapping training subsamples.

Compute: 2 runs $\times$ ~10 H100-hours = **~20 H100-hours**.

Phase 2: main study (90 runs)

The main study trains 90 GPT-2 models in a factorial design that isolates three sources of between-run variability:

- **Seed variability (30 runs).** Same training data, same architecture, different random seeds. Establishes the irreducible randomness in what a language model learns under fixed inputs.
- **Data variability (30 runs).** Same architecture, same random seed, different disjoint training subsamples (substrate chosen by the Phase 1 pilot). Measures how much of the between-run variability arises from differences in the training input — the language-model analog of how much between-child variability is driven by input differences.

- **Architecture variability (30 runs).** Same training data, varied architectural choices within the GPT-2 family (learning rate, model width, model depth). Measures how much between-model variability arises from architectural and hyperparameter choices that nominally do not change the learning problem.

For each of the 90 models, we save ~200 intermediate checkpoints across training. At each checkpoint, we measure the probability the model assigns to each of the 611 CDI words in context — yielding per-model word-learning trajectories directly comparable to the per-child vocabulary trajectories in our CDI data.

Compute: 90 runs $\times$ ~10 H100-hours per run = **~900 H100-hours**.

Per-run cost is anchored on the measured training profile for GPT-2-small on a 25M-token corpus on Sherlock A100s. 10 H100-hours per run is a conservative ceiling; at 10M tokens we expect 4–6 H100-hours per run in practice.

Storage and I/O

- **Per-run checkpoint footprint.** ~200 checkpoints $\times$ ~500 MB (GPT-2-small + optimizer state) $\approx$ 100 GB per run.
- **Project total.** 92 runs $\times$ 100 GB $\approx$ **9 TB** project scratch.
- **Per-job I/O.** Marlowe's per-node NVMe (30 TB local) covers any individual training run with significant headroom.

Software stack

PyTorch + HuggingFace transformers (matching the published Feng et al. 2026 pipeline), with HuggingFace datasets for corpus preprocessing. Post-training analysis (sigmoid fitting, between-model variability estimates, comparison to CDI data) runs in R on local hardware and is not part of the Marlowe ask. No exotic dependencies; we will request a standard Marlowe Python/PyTorch environment.

Readiness

- **Training pipeline.** Already runs end-to-end on Sherlock; 3 CHILDES seeds completed successfully under this pipeline (see model/scripts/feng_eval/ in standard-model-2).
- **Data.** CHILDES corpus already preprocessed in our group; the 10M-token chunks for the pilot will be drawn from the same 25M filtered English subset used in Feng et al. (2026). TinyDialogues (the candidate Phase 2 substrate) is publicly available from Feng et al. (2024).
- **Evaluation.** Per-word vocabulary-learning curves and the comparison to between-child variability are already implemented and validated in the parent project (model/scripts/).

- **Reproducibility.** Random seeds and training configurations will be logged for every run; per-run checkpoints, configs, and evaluation outputs will be retained for the full project lifetime.

Why Marlowe

- The 90-run main study is highly parallel across runs. Marlowe's H100 DGX nodes give the throughput needed to complete the campaign in a manageable wall-clock window (target: ~4 weeks end-to-end).
- The 9 TB project storage and high-bandwidth NVMe simplify capture of the ~18,400 intermediate checkpoints the analysis requires.